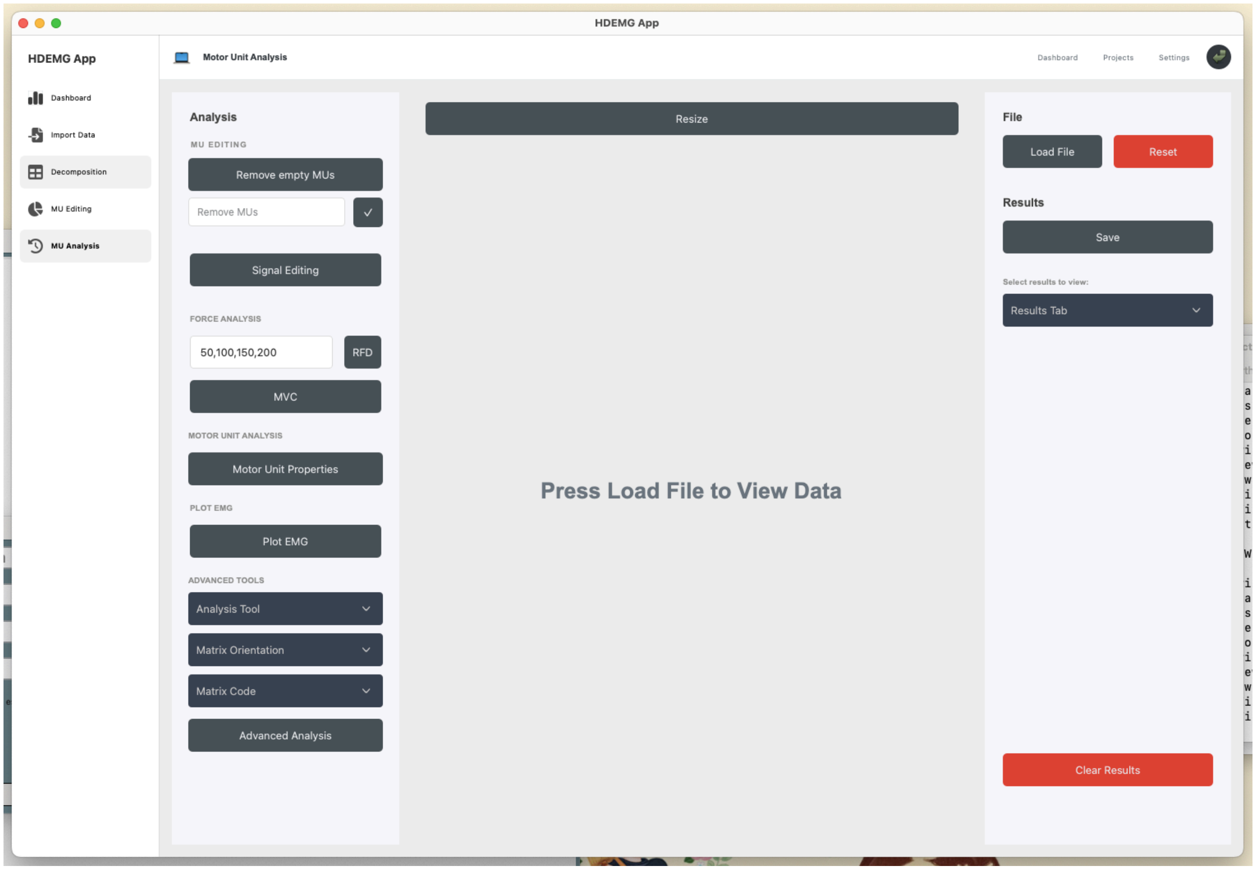
W18ABANANA - MU Analysis Tab Client Documentation

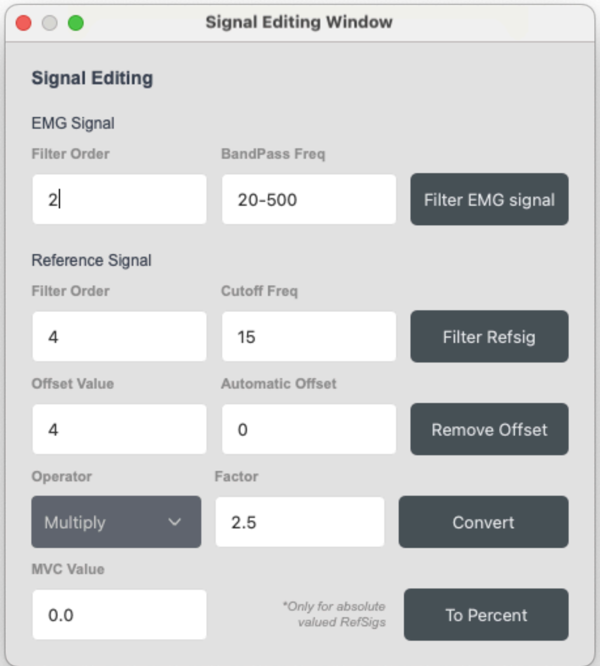
- !! NOTE: when comparing outputs to openHDEMG, please first press the “Sort MU” button, as this application sorts the MU’s by default.



Results Tab:

- Save Button: exports results in current selection as CSV
- Results Tab Drop Down: selects historical results to display and export, shows latest results at the top of the list
- Clear Results: clears all existing results, once results are cleared they cannot be recovered

Signal Editing:



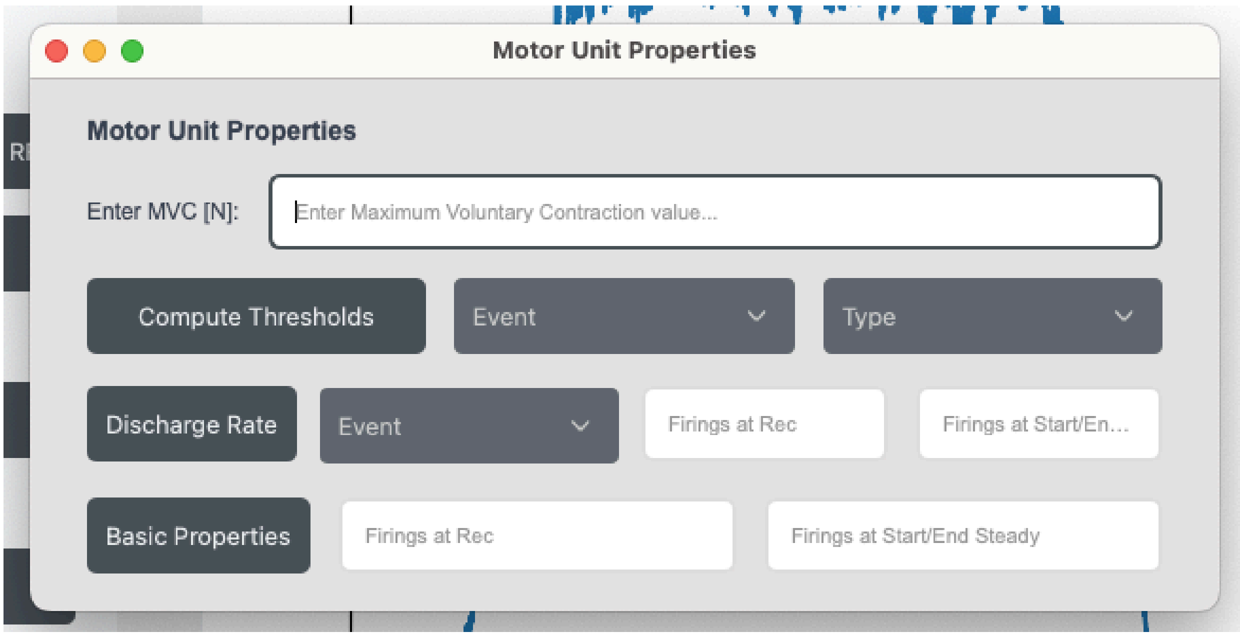
Corresponds to: https://www.giacomovalli.com/openhdemg/gui_basics/#emg-signal-filtering

Inside signal editing, there are 5 buttons the user can interact with, each button requiring the user to first fill out the text inputs in the same row. Note, all text inputs are pre-filled with default inputs. Here is an explanation of all 5 buttons:

- Filter EMG signal: Filters the EMG signal based on the filter order and BandPass Frequency and plots the result in the centre of the screen. Note, this can often look very similar to the original plot that appears in the centre when you first load a file
    - A BandPass Frequency of 20-500 filters out everything outside the range 20hz-500hz
- Filter Refsig: Low-pass filters the reference signal and removes noise based on the filter order and cutoff frequency
    - A Cutoff Frequency of 15hz filters out signals equal to and greater than 15
- Remove offset: Removes the reference signal offset and shifts the range
    - If the automatic offset is less than or equal to 0:
        - If the offset value is non-zero:
            - The offset is removed based on the offset value (e.g., an offset value of 4 will result in an offset correction by -4 in y-axis direction)
        - If the offset value is 0:
            - A prompt appears, asking the user to manually select an area, after which the offset value is computed based on the selected area

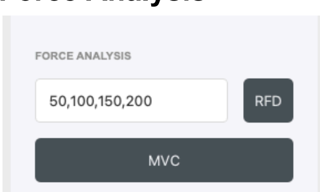
        - If the automatic offset is greater than 0:
            - The offset is automatically removed based on the number of samples passed in input
- Convert: The reference signal is multiplied/divided by the factor specified in the operator and factor text boxes respectively
- To Percent: Converts the MVC value to a percentage and divides the reference signal by the percentage

Motor Units Properties:



Corresponds to https://www.giacomovalli.com/openhdemg/gui_basics/#motor-unit-properties

Force Analysis



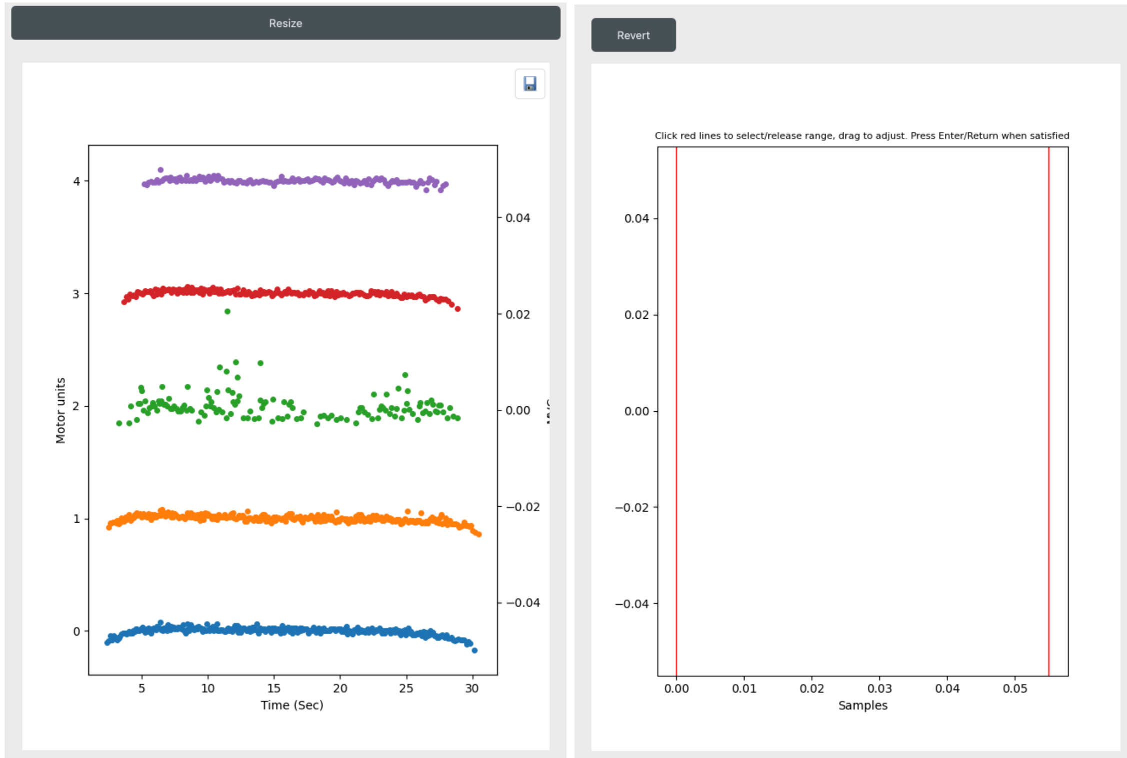
Corresponds to https://www.giacomovalli.com/openhdemg/gui_basics/#analyse-force-signal

The respective RFD values between the stated timepoint ranges (ms) in the RFD milliseconds text input are displayed in the Result Output. In example, specifying

- RFD milliseconds: 50,100,150,200

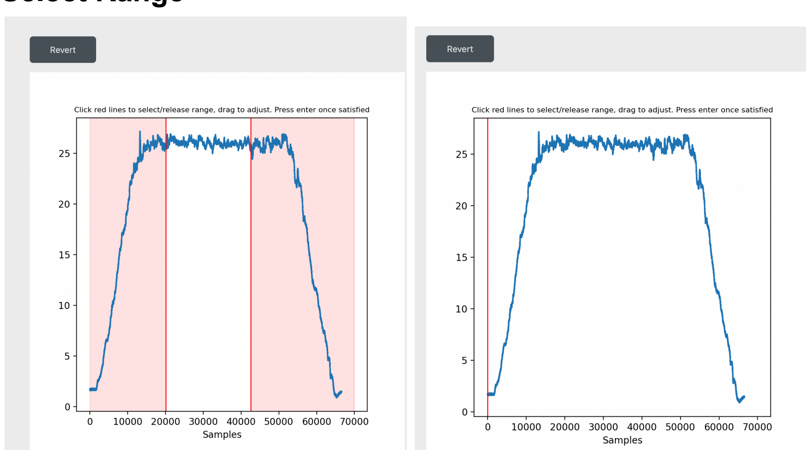
will result in RFD value calculation between the intervals 0-50ms, 0-100ms, 0-150ms and 0-200ms. You can also specify less or more values in the RFD milliseconds textbox.

Analysis Plot



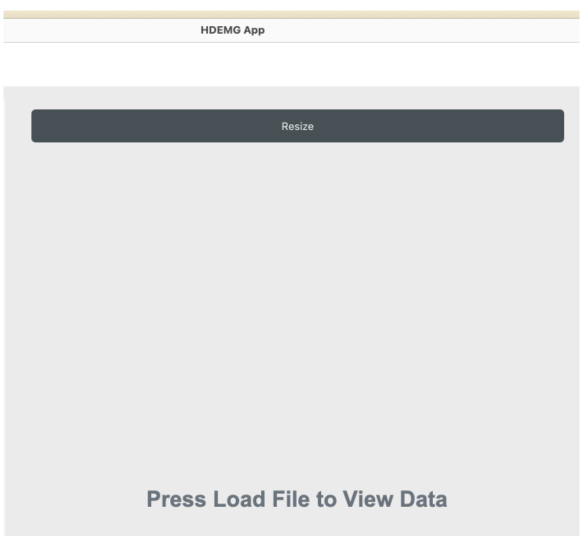
Displays plots and figures in the centre of the window. Secondary plots appear alongside a 'revert' button that appears on the top left, which, after pressing, reverts the centre plot to the last primary plot made.

Select Range



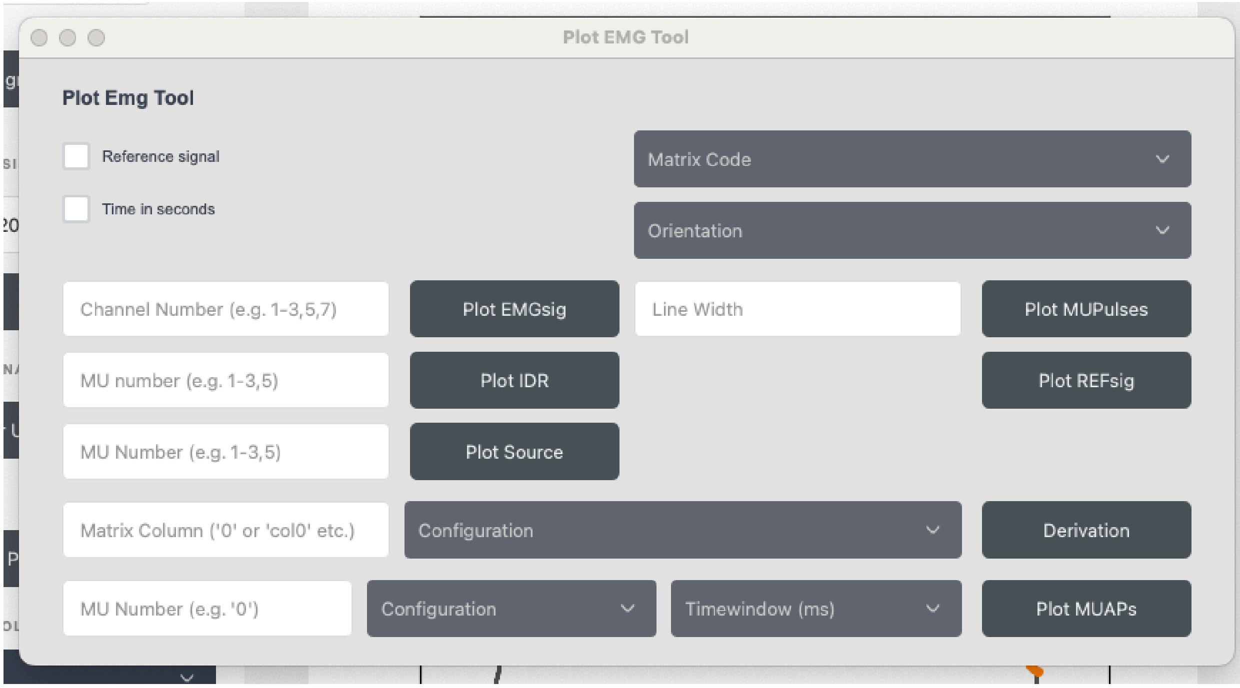
This is to replace openHDEMG's method of selecting 1 or more points

Click on a line once, move the mouse to the desired stopping position and then click again to drop the line. This can be repeated with either line as many times, just click enter when you are satisfied with its placement. Press the revert button to exit if you have changed your mind about being in this window. The lines will only move to the minimum and maximum limits of the data and the non-shaded regions emphasize the selection range.



Corresponds to https://www.giacomovalli.com/openhdemg/gui_basics/#resize-emg-file

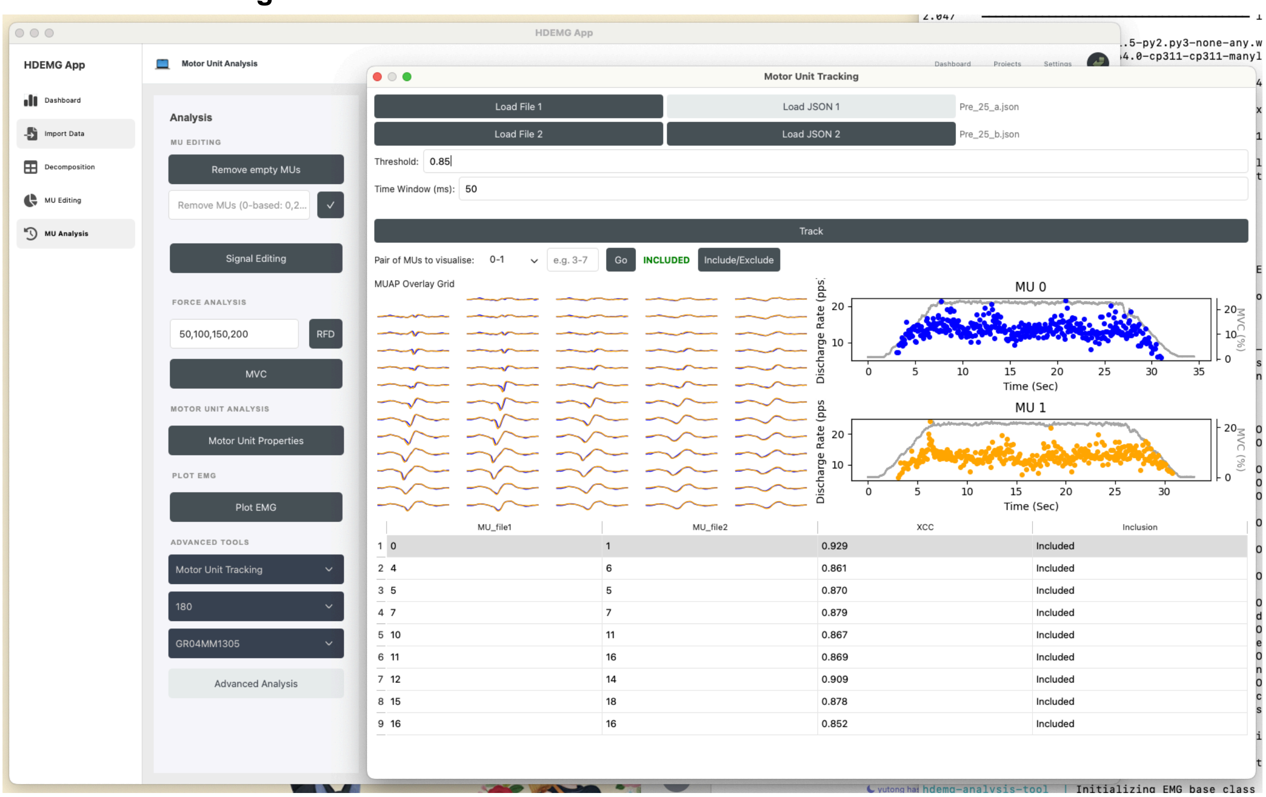
Plot EMG



Corresponds to https://www.giacomovalli.com/openhdemg/gui_basics/#plot-motor-units

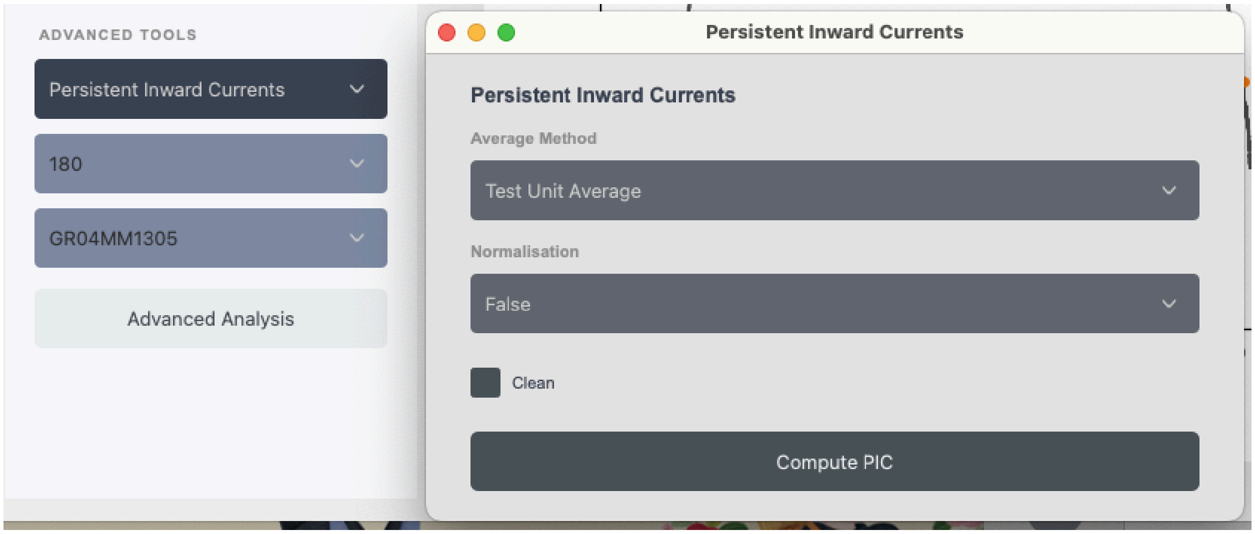
Added ability to input MU numbers as comma and dash separated values instead of drop downs

Motor Unit Tracking



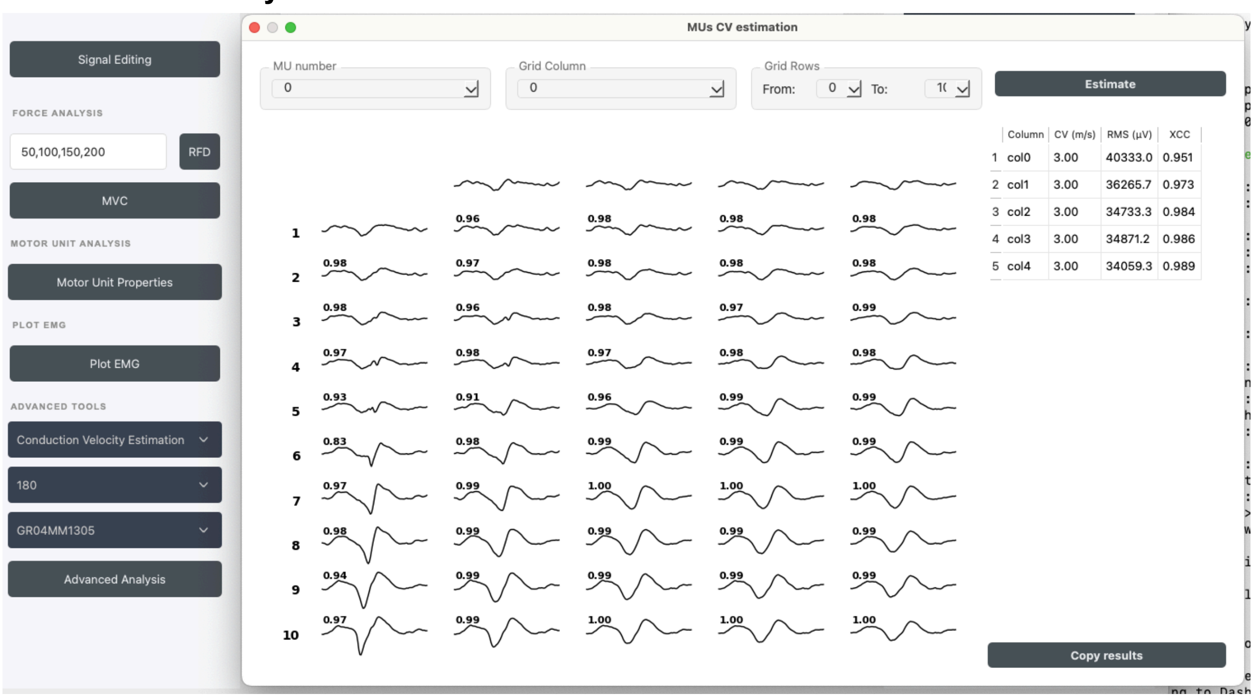
Corresponds to https://www.giacomovalli.com/openhdemg/gui_advanced/#motor-unit-tracking

Persistent Inward Currents



Corresponds to
https://www.giacomovalli.com/openhdemg/gui_advanced/#persistent-inward-currents

Conduction Velocity



Corresponds to https://www.giacomovalli.com/openhdemg/gui_advanced/#conduction-velocity
Note: the outputs of this functionality does not replicate openHDEMG values